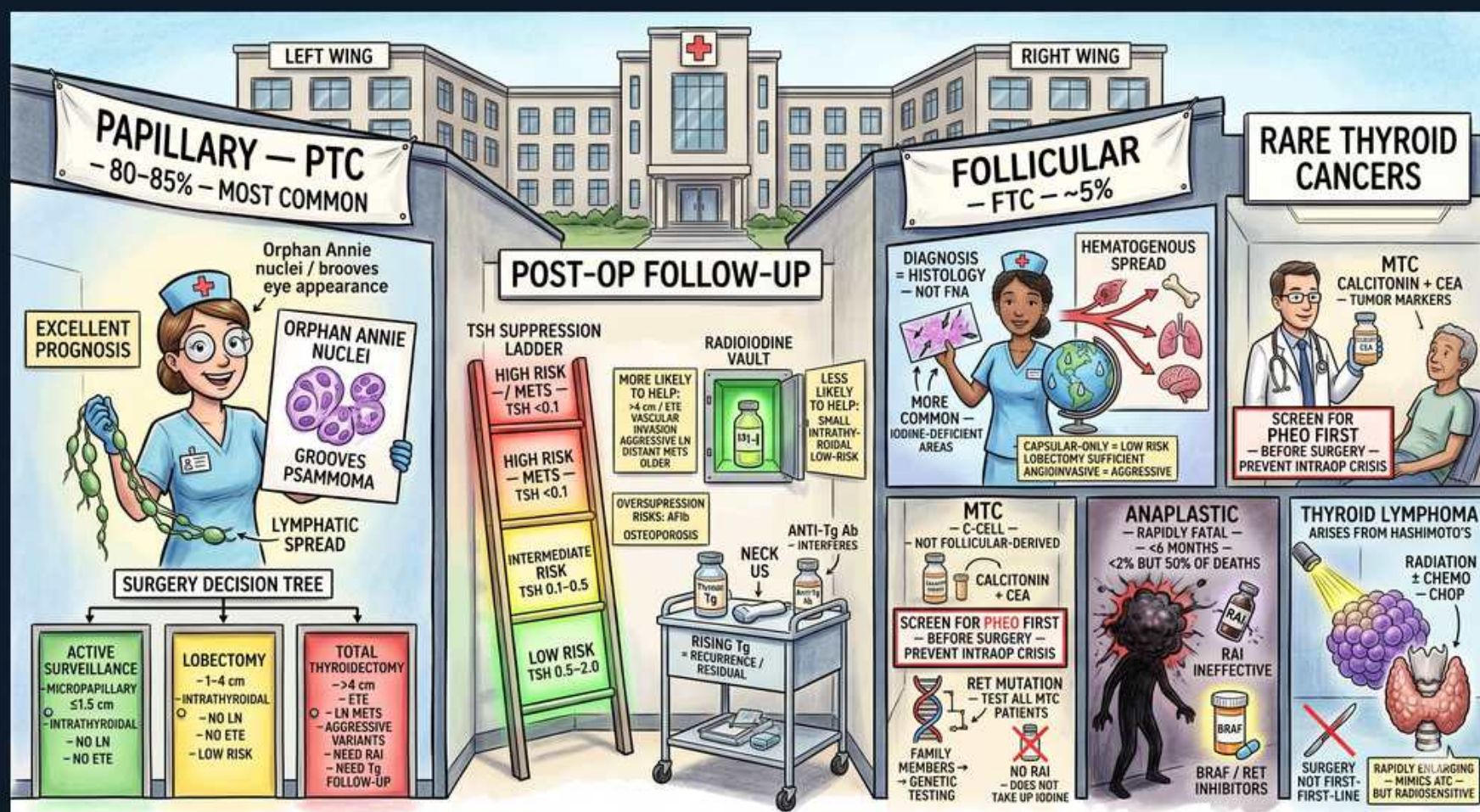


# Thyroid — Thyroid Cancers & Management





## 1. PTC most common

### WHAT YOU SEE

'PAPILLARY — PTC — 80–85% MOST COMMON'

### WHAT IT ENCODES

Papillary thyroid carcinoma is the most common thyroid malignancy (~80–85%) and a differentiated (DTC) cancer with an excellent prognosis (>95% 10-yr survival). Risk factors: childhood/neck radiation and the BRAF V600E mutation. It concentrates iodine, so it responds to TSH suppression and radioiodine.

### BOARD TRAP

Assuming the most common thyroid cancer carries a poor prognosis — wrong; PTC is the most common AND the most curable.

## 2. PTC nuclear features

### WHAT YOU SEE

'ORPHAN ANNIE NUCLEI, GROOVES, PSAMMOMA'

### WHAT IT ENCODES

PTC is a nuclear (cytologic) diagnosis: Orphan-Annie-eye nuclei (optically clear with intranuclear pseudoinclusions), nuclear grooves, and psammoma bodies (laminated calcifications). Because these are nuclear features, FNA diagnoses PTC directly — unlike follicular cancer.

### BOARD TRAP

Confusing psammoma bodies as unique to thyroid — they also appear in serous ovarian, meningioma, and mesothelioma; here the combination with Orphan-Annie nuclei is what nails PTC.

## 3. PTC = lymphatic spread

### WHAT YOU SEE

'LYMPHATIC SPREAD'

### WHAT IT ENCODES

Papillary cancer spreads via LYMPHATICS to cervical lymph nodes. Nodal metastases are common yet have relatively little impact on survival in young patients — the cancer remains highly curable. Central/lateral neck node disease guides the extent of lymph node dissection.

### BOARD TRAP

Saying PTC spreads hematogenously — wrong; that is follicular. PTC = lymphatic (paPillary → lymPh node). Mixing these up is a classic board distractor.

## 4. FTC ~5%, hematogenous

### WHAT YOU SEE

'FOLLICULAR — FTC ~5%, HEMATOGENOUS SPREAD, LOW IODINE AREAS'

### WHAT IT ENCODES

Follicular thyroid carcinoma (~5–10%) is a differentiated cancer that spreads HEMATOGENOUSLY to lung and bone (distant mets), unlike papillary's nodal spread. It typically presents as a solitary nodule and, being differentiated, still takes up radioiodine.

### BOARD TRAP

Saying FTC spreads via lymphatics or commonly to nodes — wrong; follicular = blood-borne (lung/bone) spread. (Follicular → 'Flows' through blood.)

## 5. FTC needs histology

### WHAT YOU SEE

'DIAGNOSIS = HISTOLOGY, NOT FNA'

### WHAT IT ENCODES

FTC is diagnosed only by histology showing capsular and/or vascular invasion — FNA cannot distinguish a benign follicular adenoma from carcinoma (the follicular paradox). Indeterminate follicular cytology (Bethesda IV) therefore goes to diagnostic lobectomy or molecular testing.

### BOARD TRAP

Diagnosing or excluding follicular carcinoma on FNA cytology — impossible; invasion is an architectural finding requiring the resected specimen.

## 6. FTC in iodine-deficient areas

### WHAT YOU SEE

'MORE COMMON in iodine-deficient areas'

### WHAT IT ENCODES

Follicular carcinoma is relatively more common in iodine-DEFICIENT regions, whereas papillary predominates in iodine-SUFFICIENT (or iodine-supplemented) areas. Iodine repletion of a population shifts the cancer mix toward papillary.

### BOARD TRAP

Linking iodine deficiency to papillary cancer — reversed; iodine deficiency favors follicular, iodine sufficiency favors papillary.

## 7. MTC — calcitonin + CEA

### WHAT YOU SEE

'MTC — CALCITONIN + CEA, TUMOR MARKERS'

### WHAT IT ENCODES

Medullary thyroid carcinoma arises from parafollicular C cells (neural-crest derived), so it secretes CALCITONIN (and CEA) — these are its tumor markers for diagnosis and post-op monitoring. MTC does NOT take up iodine (C cells are not follicular), so RAI and TSH suppression are ineffective; treatment is surgical ± targeted kinase inhibitors (vandetanib/cabozantinib).

### BOARD TRAP

Treating MTC with radioiodine or TSH suppression — wrong; C cells don't concentrate iodine or depend on TSH, so DTC therapy fails. Also, calcitonin (not thyroglobulin) is the MTC marker.



## 8. MTC — screen pheo first (MEN2)

### WHAT YOU SEE

'MTC — SCREEN FOR PHEO FIRST before surgery; RET mutation; test relatives'

### WHAT IT ENCODES

MTC is the thyroid cancer of MEN2 (germline RET proto-oncogene mutation): MEN2A = MTC + pheochromocytoma + primary hyperparathyroidism; MEN2B = MTC + pheo + mucosal neuromas/marfanoid habitus (no hyperparathyroidism). Before any MTC surgery, SCREEN FOR PHEOCHROMOCYTOMA (plasma/urine metanephrines) — an unrecognized pheo can trigger fatal intra-operative hypertensive crisis. Test relatives for RET; prophylactic thyroidectomy is offered to RET carriers.

### BOARD TRAP

Operating on MTC without first excluding pheochromocytoma — wrong and dangerous; you must resect/block the pheo first. Forgetting the germline RET work-up of family members is the second trap.

## 9. Anaplastic — rapidly fatal

### WHAT YOU SEE

'ANAPLASTIC — RAPIDLY FATAL, MONTHS, RAI INEFFECTIVE, BRAF/RET INHIBITORS'

### WHAT IT ENCODES

Anaplastic (undifferentiated) carcinoma occurs in the ELDERLY as a rapidly enlarging, hard, fixed neck mass with compressive symptoms (dysphagia, stridor, hoarseness); prognosis is dismal (survival in months). Being undifferentiated, it does NOT take up iodine, so RAI is useless; management is airway protection plus chemoradiation, with BRAF/MEK inhibitors (dabrafenib+trametinib) for BRAF-mutated tumors.

### BOARD TRAP

Expecting radioiodine to work or pursuing slow elective work-up — wrong; anaplastic cancer is undifferentiated (no iodine uptake) and is an airway emergency requiring urgent referral.

## 10. Lymphoma from Hashimoto

### WHAT YOU SEE

'THYROID LYMPHOMA — ARISES FROM HASHIMOTO'S, SURGERY NOT FIRST-LINE, RADIATION + CHEMO'

### WHAT IT ENCODES

Primary thyroid lymphoma (usually diffuse large B-cell or MALT type) arises on a background of long-standing Hashimoto's thyroiditis, presenting as a rapidly enlarging mass in an older patient (mimics anaplastic — biopsy distinguishes them). Treatment is chemotherapy (R-CHOP) and/or radiation, NOT surgery.

### BOARD TRAP

Taking thyroid lymphoma to surgery as first-line — wrong; lymphoma is treated medically (chemo/radiation). Also do not assume a fast-growing mass in the elderly is anaplastic — biopsy to exclude lymphoma, which is far more treatable.

## 11. Surgery decision (lobectomy vs total)

### WHAT YOU SEE

'LOBECTOMY <1cm low risk vs TOTAL THYROIDECTOMY >4cm / ETE / METS'

### WHAT IT ENCODES

Extent of surgery in DTC is risk- and size-based: lobectomy for small (<1 cm, up to ~4 cm) low-risk unifocal PTC with no extrathyroidal extension or nodes; total thyroidectomy for tumors >4 cm, extrathyroidal extension, bilateral/multifocal disease, nodal or distant metastases, or prior radiation (total thyroidectomy is also required if RAI is planned). Key surgical complications: recurrent laryngeal nerve injury (hoarseness) and hypoparathyroidism (hypocalcemia).

### BOARD TRAP

Defaulting every DTC to total thyroidectomy — wrong; small low-risk papillary cancers are adequately treated by lobectomy (or even active surveillance), avoiding lifelong replacement and surgical risks.

## 12. Active surveillance micropapillary

### WHAT YOU SEE

'ACTIVE SURVEILLANCE micropapillary <1cm'

### WHAT IT ENCODES

Papillary microcarcinoma (<1 cm), low-risk and without nodal/extrathyroidal disease, may be managed with active surveillance (serial ultrasound) rather than immediate surgery, given its indolent behavior and excellent outcomes.

### BOARD TRAP

Insisting all thyroid cancers need immediate resection — wrong; low-risk micropapillary PTC can be safely observed.

## 13. TSH suppression ladder

### WHAT YOU SEE

'TSH SUPPRESSION LADDER — HIGH RISK TSH <0.1, LOW RISK 0.5–2; OVERSUPPRESSION → AFib/OSTEOPOROSIS'

### WHAT IT ENCODES

After surgery for DTC, levothyroxine is dosed to SUPPRESS TSH because TSH is a growth factor for differentiated thyroid cancer cells. Target by recurrence risk: high-risk TSH <0.1, intermediate ~0.1–0.5, low-risk ~0.5–2 mIU/L. Suppression is relaxed once the patient is disease-free.

### BOARD TRAP

Over-suppressing TSH long-term in low-risk patients — wrong; excessive suppression causes atrial fibrillation and osteoporosis, so the target is risk-stratified and de-escalated over time. (Note: TSH suppression only helps DTC — it does nothing for medullary or anaplastic cancer.)

## 14. RAI + Tg/anti-Tg follow-up

### WHAT YOU SEE

'RADIOIODINE VAULT; THYROGLOBULIN / ANTI-Tg Ab, NECK US for recurrence/residual'

### WHAT IT ENCODES

Radioactive iodine (I-131) ablates residual/microscopic differentiated cancer after total thyroidectomy in higher-risk DTC, and enables surveillance. Follow-up uses serum THYROGLOBULIN (a tumor marker after the normal gland is removed — a rising Tg signals recurrence) plus neck ultrasound; whole-body RAI scans for higher-risk cases. Recombinant TSH or thyroid hormone withdrawal raises TSH to stimulate iodine uptake before RAI/scanning.

### BOARD TRAP

Trusting a 'normal' thyroglobulin when anti-thyroglobulin antibodies are present — wrong; anti-Tg antibodies falsely LOWER measured Tg, masking recurrence, so anti-Tg must always be measured alongside Tg. Also: RAI is useless in medullary/anaplastic cancer (no iodine uptake).